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Intelligent application of large language model to life cycle assessment methodology

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HIGHLIGHTS

- Systematic combination of LLMs and LCA breaks through the traditional LCA bottleneck.
- RAG-Based LCA domain LLMs reduce knowledge threshold achieving QA BERTScore of 0.85.
- Utilizing LLMs to enhance LCI database retrieval by Text2SQL, attaining EX of 0.97.
- Enabling carbon footprint report auto-generation through LLM code-interpreter agent.

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ABSTRACT

Traditional life cycle assessment (LCA) methods face limitations such as high dependence on manual labor and expertise, difficulties in retrieving background databases, and low efficiency in manually writing carbon footprint reports, which can be addressed by large language models (LLMs). However, current research focused on a limited number of specific areas. This paper proposed a ubiquitous method for LLMs enhanced LCA research including knowledge summarization, data retrieval, and report generation. Firstly, a professional question-answering (QA) tool with knowledge summarization in LCA was realized utilizing knowledge base and retrieval-augmented generation method to alleviate hallucinations. Then, a natural language retrieval method for querying life cycle inventory (LCI) databases was developed by enhancing the LLMs with few-shot Chain of Thought and Chain of Code in Text2SQL. Furthermore, with code-interpreter, the LLM agent was able to auto-generate carbon footprint reports following a template. Finally, multiple experiments were performed to evaluate the performance of the different methods. The QA performance was evaluated using a QA dataset, achieving the BERTScore of 0.85, which demonstrates its capability in knowledge acquisition and summarization. The ability of database retrieval was validated by execution accuracy, and the highest value 0.9692 was obtained after augmented prompt engineering. The accuracy and completeness of the report were assessed through five dimensions, with the proposed model performing best in four of them. This paper provides a theoretical basis and experimental verification for the intelligent transformation of the LCA method. Future research will focus on enhancing interpretability of LLMs, expanding LCA-specific datasets, and improving intelligent LCI management to advance sustainable development.

1. Introduction

As an important tool for carbon footprint accounting, LCA is used to characterize the complex, multidimensional and potential impacts of human activities, which can provide quantitative information to help make better decisions (Liu et al., 2024, 2022). The LCA method framework consists of four stages as Fig. 1 shows: goal and scope definition, inventory analysis, impact assessment and interpretation (Standardization, 2006). The goal and scope definition involves the identification of the specific research question, intended application and target audience, as well as the establishment of system boundaries and functional units. The inventory analysis is a key step in LCA where all input and output data from the full life cycle of the product system are

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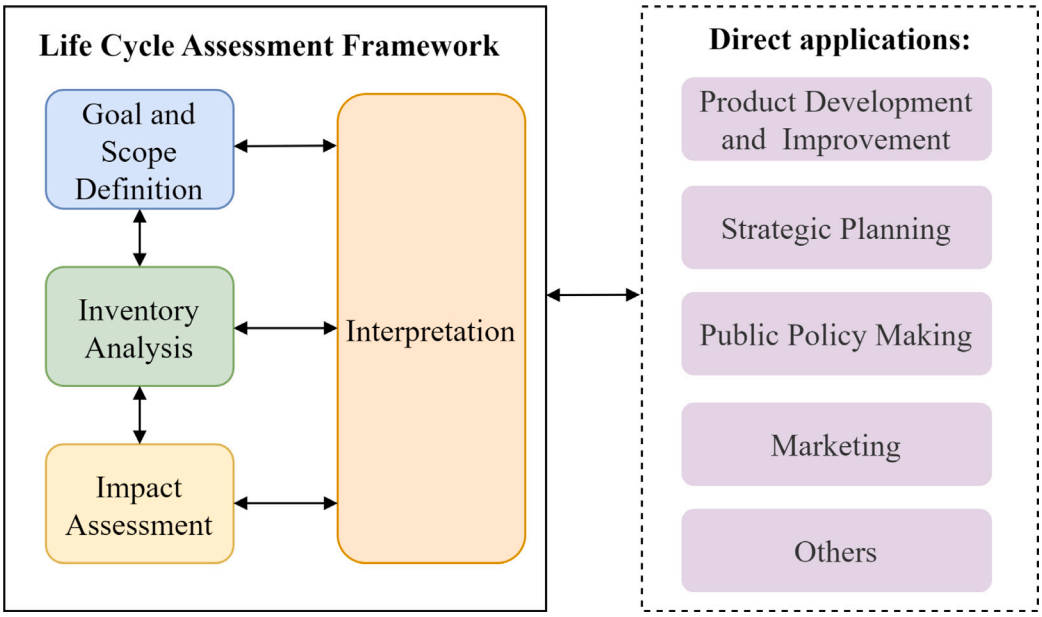


Fig. 1. Life cycle assessment framework.

collected and summarized. The impact assessment acts as a bridge between the inventory and the impacts by qualitatively or quantitatively assessing the resource consumption and pollution emissions. After that, data from the impact assessment are thoroughly analyzed in the interpretation phase, which aims to identify key factors and potential hotspots in the environmental impacts of the product or service across its life cycle. To promote green and low-carbon development, non-specialized LCA practitioners are increasingly involved in carbon footprint accounting. However, the establishment of a full LCA process is complex and detailed. There is currently a lack of effective tools to help non-experts understand LCA concepts (Mudersbach et al., 2023). Furthermore, traditional LCA research relies on huge databases, and the process of database querying and matching is labor-intensive and time-consuming (Schneider et al., 2023). In the life cycle impact assessment (LCIA) stage, how to develop targeted carbon reduction methods and generate high-quality carbon footprint reports for different products are also important issues (Harding, 2013).

Advancements in artificial intelligence (AI) and machine learning (ML) have enhanced LCA research (Liu et al., 2024), with applications in generating LCI with ML (Romeiko et al., 2024), calculating characterization factors (Ghoroghi et al., 2022), estimating life cycle impacts (Algren et al., 2021) and supporting interpretation (Zhu et al., 2020). However, these studies still face limitations in the efficiency of LCA knowledge understanding, the challenges in inventory collection and matching, and the lack of automation in report generation (Romeiko et al., 2024). LLMs, leveraging contextual learning and multi-step execution, offer solutions and have been applied in finance (Li et al., 2023b), law (Cui et al., 2023), and healthcare (Ghosh et al., 2024). While the theoretical potential of applying LLMs to LCA has been explored, challenges and risks remain (Preuss et al., 2024). Subsequent studies have leveraged LLMs' text understanding and reasoning abilities in LCA. For example, GPT-3.5 has been used to read and summarize large volumes of machine learning literature, helping to fill LCI data gaps and reduce literature review time (Chen et al., 2024). LLMs mitigate the challenges of analyzing corporate sustainability reports that are information-intensive and expensive to analyze by delivering comprehensible insights from the reports (Ni et al., 2023). Furthermore, LLMs for LCA data collection and matching have also been explored. LLMs can generate and describe LCI for a limited number of products based on pre-trained data (Cornago et al., 2023). However, the limited amount of data constrains the capability. Tu et al. (2024) constructed an open LCI database by processing existing sources and collecting public LCA results. LLMs assist in rapidly identifying emission inventories, while deep learning models generate and match activity data with impact factors (Luo et al., 2023). Additionally, the study leverages LLMs with retrieval-augmented generation (RAG) techniques to manage LCI, which helps scientists quickly locate the required data (Bekavac and Volk, 2024). Nevertheless, the existing research has not fully addressed the challenges in LCA. Basic LLMs struggle with hallucinations, outdated information, and lack of transparency in understanding LCA domain knowledge. Specialized LLMs for aggregated LCA learning are needed to improve access to domain knowledge. In LCI database retrieval, the RAG technique relies on semantic similarity but falters with detailed user conditions. Additionally, the potential of LLMs to manage both structured and unstructured data remains underexplored, with limited studies on utilizing their text generation and data analysis capabilities across the LCA process.

To address the challenges, this study aims to improve LCA efficiency by leveraging LLMs, specifically through creating a professional QA tool, facilitating LCI database access by natural language (NL), and automating LCIA report generation with high accuracy. Currently, the application of RAG in the financial and legal fields is relatively mature, Text2SQL in database retrieval has shown good results, and LLMs have the potential to automatically generate carbon footprint reports with their powerful text generation capability. This paper proposed an approach to develop the full LCA research by integrating LLM domain specialization techniques, as shown in Fig. 2. First, to overcome the knowledge-intensive nature of LCA, we build a professional QA service based on RAG technology to understand and acquire knowledge. Authoritative LCA articles are collected to build the knowledge base. Then, an LLM-enhanced method for retrieval of vector data is created by combining Text2SQL and prompt engineering, which enables efficient access from NL to the LCI database. Furthermore, given LLMs' ability to understand and process text, the automation of carbon footprint report generation with LCIA data analysis is achieved by the assistance of code-interpreter agent. Finally, we design different metrics to assess the above components separately. Our contributions include three aspects:

- This paper proposes a generalized method for applying LLMs to the LCA domain, which realizes the purpose of enhancing the intelligent research of LCA.
- Based on the domain-specific technology, we implement three applications of knowledge summarization, data retrieval and report generation to make LCA process more efficient.

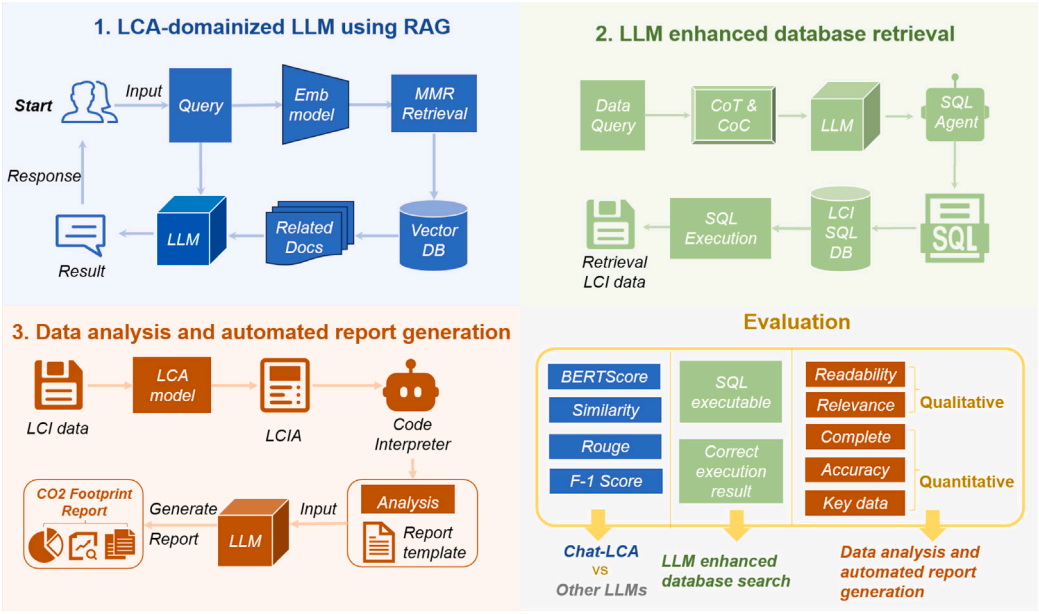


Fig. 2. Research method design and evaluation framework.

- Assessment datasets are constructed to enable targeted evaluations of various application experiments, incorporating both qualitative and quantitative methods.

The remain of the paper is organized as following parts. Section 2 of this paper presents the methods, which includes the construction of a domain-specific LLM for LCA using RAG, the retrieval for LCI databases using LLMs and Text2SQL with few-shot Chain of Thought (CoT) and Chain of Code (CoC), and the automation of LCIA report generation with data analysis based on code-interpreter agent. Sections 3 and 4 present the experimental results and evaluation respectively. Both quantitative and qualitative evaluation experiments are conducted on the three research contents. Section 5 summarizes the advantages and potential for improvement of the research method, and discusses the future direction for advancing the application of LLMs in LCA.

2. Method

The LLM domain specialization methods include three approaches: external augmentation, prompt crafting, and parameter-efficient fine-tuning (Ling et al., 2023). In the study of the professional QA tool Chat-LCA, the RAG technique is applied to enable the LLMs to acquire LCA domain knowledge and facilitate intelligent QA in LCA (Khan et al., 2025; Hang et al., 2025). Based on the theory and method of Text2SQL, LLMs and augmented prompt with CoT (Wei et al., 2022) and CoC (Li et al., 2023a) are adopted to realize the retrieval of LCI database by NL. The code-interpreter agent in LLMs is combined to achieve automated report generation for LCIA with data analysis.

2.1. Construction of LCA-domainized LLM based on RAG

The RAG is employed in Chat-LCA, which enhances the accuracy and credibility of the model by integrating knowledge from external databases. RAG is especially beneficial for knowledge-intensive tasks, as it allows continuous knowledge updates and utilization of domain-specific information. It overcomes the LLMs' limitations and provides a more reliable solution for handling domain-specific tasks. RAG consists of three steps shown in Fig. 3: knowledge base building, knowledge retrieval, and LLM's answer generation (Lewis et al., 2020). Compared to prompt engineering, RAG leverages external knowledge and provides high interpretability of responses, enhancing the accuracy of professional QA. Compared to parameter fine-tuning, which requires more time and computational resources, RAG offers timely data updates, security and privacy management, and reduced training costs. Therefore, RAG is chosen to implement Chat-LCA.

2.1.1. LCA domain knowledge base building

A professional LCA knowledge base requires authoritative data collection, with duplicate and invalid entries removed to ensure quality. For unstructured data, semantic integrity is maintained through content splitting and vectorization, enabling efficient vector retrieval.

In the RAG framework, the knowledge base is stored in a vector database. Since the original data is in text format, a text embedding model is used to convert the acquired scientific paper text into vectors. Text embedding model involves projecting text into a continuous vector space, aiming to preserve the semantics and similarity between texts (Patil et al., 2023).

Massive Text Embedding Benchmark (MTEB) provides a diverse benchmark for text embedding models to evaluate the usability of text embedding models in different application cases (Muennighoff et al., 2022). A text embedding model suitable for the application scenario of this paper can be selected based on MTEB.

2.1.2. Efficient retrieval of knowledge base

The text vectors generated through text embedding are stored in a local database. To enhance the efficiency of retrieval, the Maximum Marginal Relevance (MMR) search method is employed for retrieving input content based on relevance (Sarmah et al., 2023). The primary objective of MMR is to reduce redundancy while maintaining query relevance, which is achieved by reordering the retrieved documents. During the retrieval process, the MMR score for each query sentence is calculated following the formula (1):

$$MMR(Q_i) = \lambda \times Sim1(Q_i, D) - (1-\lambda) \times Sim2(Q_i, Summ) \quad (1)$$

where D is the text vector in the database, $Summ$ is the set of texts that have been retrieved, λ is a parameter to balance relevance and redundancy ranging from 0 to 1, $Sim1(Q_i, D)$ is a similarity function to measure the similarity between the query sentence Q_i and the text vector D in the database representing the relevance, $Sim2(Q_i, Summ)$ is another similarity function that measures the similarity between the query sentence Q_i and the sentence vector $Summ$ in the retrieved text set to avoid redundancy.

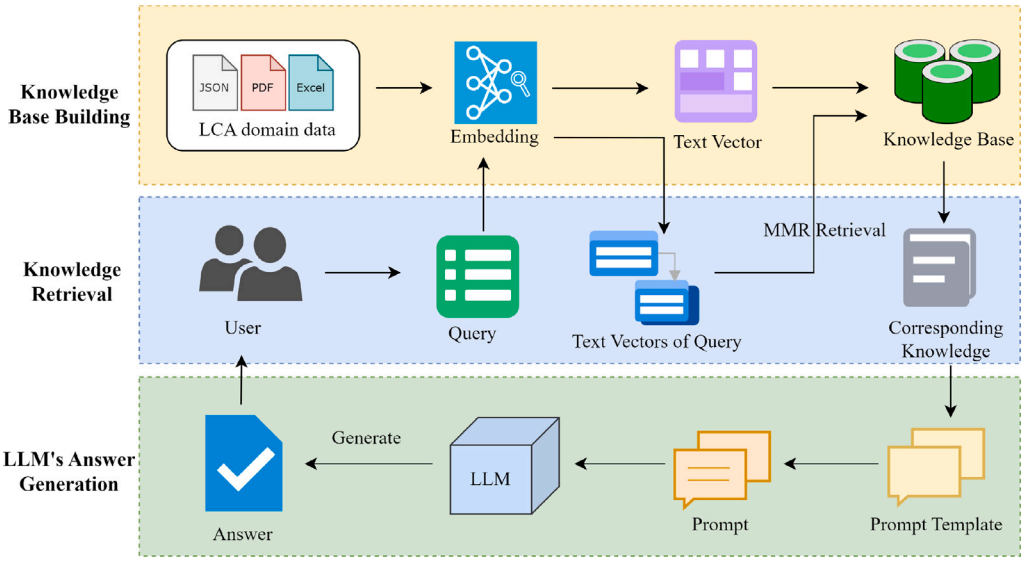


Fig. 3. RAG framework.

MMR sequentially selected high-scoring text vectors for the retrieval set until the predefined length was reached. By considering similarity between the query sentence and both the document and retrieved text, MMR reduced duplication and redundancy. It balanced relevance and diversity, ensuring pertinent yet varied content. Parameter adjustments prioritized summary diversity without sacrificing relevance. The simplicity and effectiveness of MMR made it ideal for efficient knowledge database retrieval, leading to its adoption for efficient retrieval.

2.1.3. Constructing LCA-domainized LLM using RAG

After constructing the LCA domain knowledge base and selecting an efficient retrieval mechanism, the following step is to enable the LLMs to learn specialized LCA knowledge and provide accurate responses to user queries. The construction process is based on RAG, which utilizes the knowledge base to enhance the reasoning and generative capabilities of the LLMs in the domain. The entire process includes the following key steps as Fig. 3 shows:

- When a user initiates a query for the LCA domain, the text embedding model transforms the query into text vectors.
- Based on the query text vectors, the system retrieves relevant information in the knowledge base of the LCA domain. Using MMR retrieval mechanism, the content related to the query is extracted from the domain knowledge base. Then the content will be organized into a new prompt and sent to the LLMs for processing.
- LLMs integrate the retrieved domain knowledge with its own pre-training data, and generate the final answer by taking into account the background and context of the retrieval.

To overcome LLMs' context length limits, long texts from the LCA knowledge base are segmented, vectorized, and stored in a vector database. Leveraging the MMR algorithm, the method selects text segments with high query relevance ($Sim1$) and low redundancy ($Sim2$), forming concise prompts for the LLM. This ensures accurate retrieval without directly processing long texts, effectively bypassing context constraints.

2.1.4. Professional QA assessment dataset

To build a dataset with LCA-specific knowledge for evaluating ChatLCA's answer accuracy, we collected authoritative LCA papers and used Qwen-max to generate QA pairs in Chinese, as shown in Fig. 4. Qwen-max initially created 5078 QA pairs ($Dataset_0$) from LCA documents, but some were incomplete or cluttered. Then, the qualified data in $Dataset_0$ were labeled as Positive, while the unqualified data were labeled as Negative. An LCA expert annotated and partitioned $Dataset_0$ into qualified data with 3933 QA pairs ($Dataset_1$) and unqualified data with 1145 QA pairs ($Dataset_2$). To verify the accuracy of the initial annotation, two additional LCA experts independently tested samples of 200 pairs from $Dataset_1$ and 100 from $Dataset_2$ to calculate the True Positive (TP) in $Dataset_1$ and the True Negative (TN) in $Dataset_2$. Cohen's Kappa score was employed to measure the inter-annotator consistency for the samples. The Cohen's Kappa score was calculated by the formula (2):

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{2}$$

where p_o represents the observed agreement probability between annotators, and p_e is the expected agreement probability under random conditions. The κ values range from $[-1, 1]$, where 1 signifies perfect, non-random agreement, 0 indicates agreement equivalent to chance, and -1 suggests agreement significantly below chance expectations.

The consistency test for sampled QA pairs produced Cohen's Kappa scores of 0.6426 for $Dataset_1$ and 0.6398 for $Dataset_2$, indicating substantial annotator agreement. Discrepancies were resolved using Kimi,[1] with the majority result as the final outcome. In $Dataset_1$, 187 samples (93.5%) were identified as TP, indicating qualified data, while in $Dataset_2$, 91 samples (91.0%) were classified as TN, confirming unqualified data. These results substantiate the accuracy of the initial screening process.

We classified the QA dataset using the Industrial Classification of National Economic Activities (CS (Chinese Standards), 2017) shown in Fig. 5, identifying the top ten industry categories, totaling 3018 entries. Abbreviations for the ten industry categories appeared in Appendix B. Approximately 50% of the data focused on LCA theory, ecological protection, and environmental governance. Research and experimental development and construction industries both occupied about 10%, with the rest distributed across non-metallic mineral products, chemical manufacturing, and other sectors.

2.2. Retrieval of LCI database based on LLM

The retrieval of LCI databases under specific user-defined requirements is grounded in the fundamental principles of Text2SQL and enhanced through few-shot learning techniques. These methods enable LLMs to facilitate fast and accurate retrieval between NL queries and LCI databases.

[1] https://www.kimi.com/.

Prompt:
You are a Data Analysis Associate specializing in the field of Life Cycle Analysis (LCA). You have extensive experience and knowledge in the field of LCA, including goal and scope definition, data inventory collection and analysis, life cycle impact assessment, results analysis, and policy suggestions.

Instruction:
Please ask 10 questions and corresponding answers based on the document below:
The output of the 10 questions are 10 strings, starting with "#question1:",ending with "?". The output of 10 answers are 10 strings, starting with "@answer1:", ending with ".".

Questions:
#question 1: What is the LCA-CEE model used for?
#question 2: What is truncation error in life cycle assessment? ……

Answers:
@Answer 1: The LCA-CEE model is used to assess the combined carbon footprint, environmental and economic impacts of a wastewater treatment plant over its full lifecycle.
@Answer 2: Truncation error arises from counting only the parts that have a major impact on the study and ignoring the minor parts, resulting in an incomplete product life cycle. ……

Fig. 4. QA dataset construction.

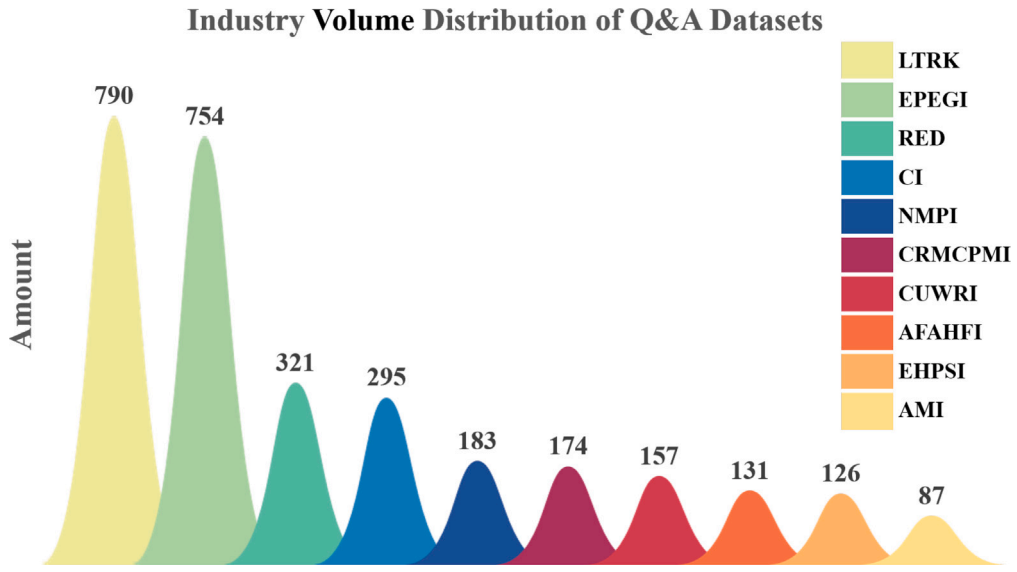


Fig. 5. Industry distribution of QA datasets.

2.2.1. Text2SQL basic methods

LLMs are excellent at processing textual information but cannot directly access databases. To overcome the limitation, NL is transformed into SQL to enable indirect database access. With the expansion of training corpora, the code generation capability of LLMs gains recognition among researchers (Ouyang et al., 2023). This advancement enhances the efficiency of converting NL into structured code, such as SQL.

Text2SQL is an essential task in natural language processing (NLP), which focuses on translating NL questions into SQL queries, thereby bridging the gap between non-expert users and database interaction (Shi et al., 2024). This approach significantly improves data processing efficiency and broadens its applicability across various scenarios (Gao et al., 2023).

This paper builds upon Text2SQL theory to enhance the accuracy of data retrieval. LLMs improve text comprehension and standardize code output. The SQL queries generated by LLMs connect the interaction between NL and the LCI database. The primary methods of applying LLMs to Text2SQL tasks include prompt engineering and parameter-efficient fine-tuning.

2.2.2. LCI database retrieval based on few-shot CoT and CoC

The basic framework of the prompt engineering method consists of NL question and the database's organizational mode. The key lies in guiding LLMs to generate accurate SQL queries through precise prompts. To achieve this, a series of templates or prompts should be designed to help the LLMs understand the user's query intent and convert it into the corresponding SQL queries. For instance, by providing the database structure, field names, data types, and example queries, LLMs can better grasp the question context and generate more precise SQL queries.

A hybrid prompt engineering method, integrating few-shot CoT and CoC, is implemented. Few-shot learning allows LLMs to adapt to diverse database structures and queries with few examples (Ling et al., 2023). In the Text2SQL task, prompts encompass instructions, context, database schema, and SQL output format. CoT enables step-by-step problem decomposition, while CoC provides clear computational logic via intermediate code or pseudocode, ensuring precise semantic parsing and structured output. LLMs support rapid retrieval from the LCI database, as shown in Fig. 6. NL queries from users are incorporated into prompt templates for the LLM-SQL agent, which produces and executes SQL queries to fetch the necessary data.

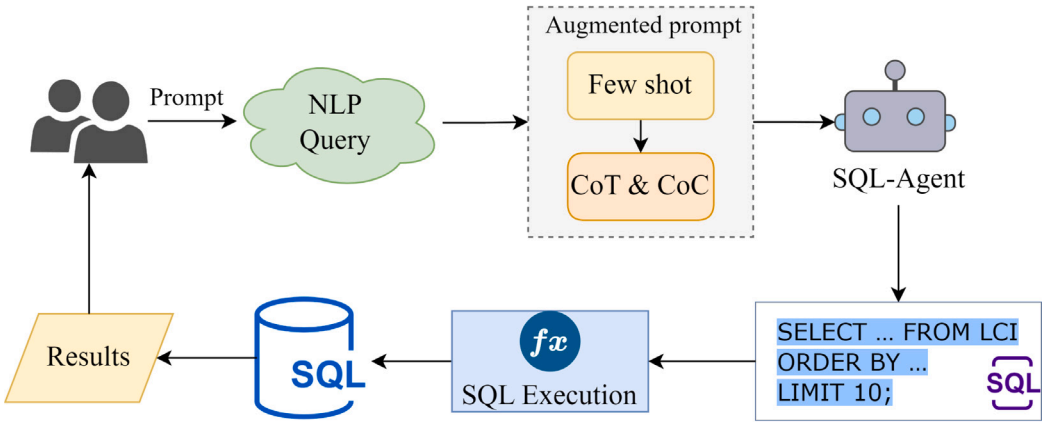


Fig. 6. LLMs' enhanced LCI database retrieval.

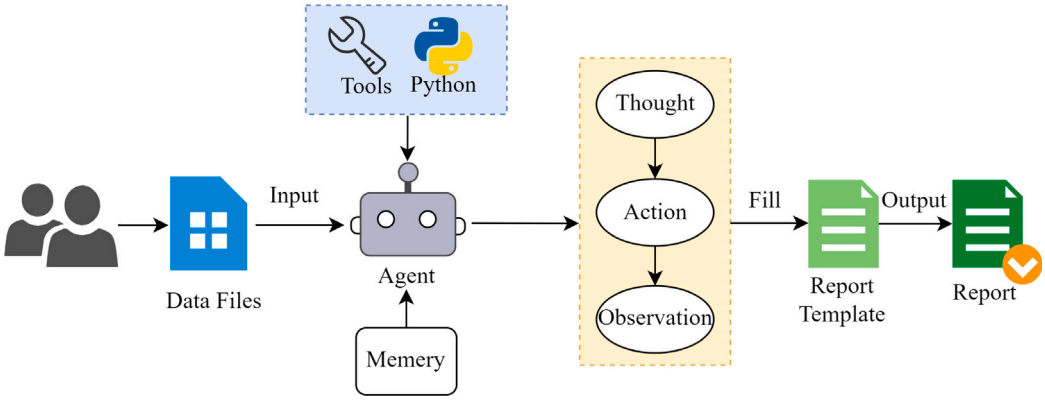


Fig. 7. LCA data analysis and report generation automation flowchart.

2.3. Automation of LCA report generation

Leveraging deep learning techniques, LLMs have developed advanced capabilities in pattern recognition, relational inference, and prediction to perform complex analysis and reasoning on input data without the need for explicit programming instructions (Hadi et al., 2023). The application of LLMs to assist with data analysis in LCA research is explored. Following data analysis, we harness the LLMs' ability to comprehend and process textual data to learn professional report templates in the LCA field. The LLMs then interactively generates reports realizing the user's specific needs.

2.3.1. Application of LLMs in LCA data analysis

LLMs' capabilities in mathematical reasoning and data processing remain imperfect. The code-interpreter agent enables data analysis through code creation and execution, which supports logical natural language reasoning, generating and executing code step-by-step. Execution outputs feed back to LLMs for result analysis and interpretation. By integrating code generation and execution into NLP, limitations in mathematical reasoning and data processing are mitigated. This advancement markedly enhances LLMs' potential for complex data analysis tasks (Zhou et al., 2023).

The steps for implementing LCA domain data analysis using LLMs are showed in Fig. 7. The user provides a file for analysis, and the code-interpreter processes it by reasoning, writing, and executing code based on instructions. Visualization requirements can also be provided to standardize the data analysis interactively and then the results are generated. A zero-shot learning approach is adopted, where the prompt includes only the task description (Ling et al., 2023). The prompt is designed to guide the LLMs in analyzing various data files, which informs the LLMs about the meaning of each data column and specifies the functional units in LCA. Following the above steps, the LLMs can perform the tasks as required.

2.3.2. Automated LLM-based report generation

Language modeling is the most basic capability of LLMs, aiming at predicting the next token based on previous tokens, and focusing on language understanding and generation abilities (Zhao et al., 2023). Accordingly, automated report generation uses the LLMs to understand the report structure and filling it with existing information.

Data analysis results serve as the basis for report generation. The report's objective and scope are defined, and a template aligned with product LCA reporting requirements is specified, ensuring LLMs comprehend and retain it. LLMs fill the template and produce customized text based on input data characteristics and values. Visualizations from the data analysis, paired with suitable headings and annotations, are integrated into the report. Missing information is identified by LLMs for manual addition through interactive processes. The report's formatting is verified for standard compliance, with layout and style adjusted for readability and consistency.

3. Experiments

In order to verify the feasibility of the above method, this paper conducts experiments on professional QA tool Chat-LCA, data retrieval and report generation based on LCA domain data.[2] Chat-LCA offers an effective pathway for knowledge acquisition in LCA, allowing users to rapidly access relevant background information and develop a clear understanding of inventory data query requirements. Through the integrated data query module, users can accurately retrieve the necessary

[2] The code and dataset are available: https://github.com/forTomorrow-0/LCA-QA-dataset.git.

Table 1
RAG framework parameter.

Parameter	
Similarity function	Cosine similarity
MMR λ	Default
Top-k	10
Temperature	Default
Chunk_size	1000
Chunk_overlap	200

data from the LCI database. The knowledge acquisition and data querying jointly provide comprehensive support for the LCA modeling process. Upon completion of the modeling, the report generation module automatically produces structured and standardized carbon footprint assessment reports based on the LCIA results.

3.1. LCA domain professional QA

In this paper, the database was constructed primarily using scientific and technical papers, and the data source was China Knowledge Network. By searching the keywords "life cycle assessment" and "LCA", academic journals and conference papers were selected as the target literature types. This paper identifies a preliminary collection of domain literature by relevance to keywords. After manual verification, 600 high-quality scientific and technical papers were ultimately selected, and all the papers were stored in PDF format.

During the experiment, referring to the MTEB list, the top-ranked Chinese text embedding model acge_text_embedding[3] was selected and employed as the text embedding model. The scientific and technical papers are read, processed and embedded to generate the text vectors, which are stored in a local vector database as the LCA professional domain knowledge base. Chroma,[4] a new AI-native open-source embedded database, is lightweight and easy to use. In this paper, Chroma is used to encapsulate the knowledge base vectors and saved them to disk. MMR is applied to retrieve the vector database and support subsequent LLMs' retrieval queries. The RAG parameters are shown in Table 1.

An interactive visualization interface with a QA function, built on RAG technology and the Streamlit framework, is depicted in Fig. 8. The image demonstrates that our approach is equally applicable to English conversations. Real-time interaction with Chat-LCA in specialized LCA domains delivers immediate, field-specific responses. Document uploads expand the knowledge base, enabling adaptation to new information and ensuring accuracy and currency. Conversation history is retained, allowing review of past interactions and providing context for precise answers. In multi-turn dialogues, prior interactions are recalled to maintain consistency and relevance.

3.2. Retrieval of LCI database

This paper employed the few-shot learning method and the prompt template designed in Section 2.2, along with API calls to the LLMs. Docker was used to deploy the experimental database locally, which included several data tables for testing. The prompt input format used in this study is shown in Appendix C. First, the table structure of the experimental database is retrieved and provided to the LLMs. Then, the LLMs are instructed to memorize this structure and generate SQL queries that meet the specifications based on user questions. Finally, a program is written to automatically execute the SQL statements and enable retrieval to the database. The results from executing this process are shown in Table 2, demonstrating the accuracy of the SQL statements and execution outcomes.

Table 2
NL question transformation SQL query process.

Process	Content
NL question	Find the number of records in the tb_process table that are created between 2022-01-01 and 2023-01-01.
SQL query	`SELECT COUNT(*) FROM tb_process WHERE create_time BETWEEN '2022-01-01' AND '2023-01-01'`
Execution results	[(21238,)]

To evaluate the LCI database retrieval method enhanced by prompt engineering, background logs from the GIS-LCA[5] database were used to extract SQL queries. The SQL log files are stored in MyBatis format, with SQL queries separated from their parameters. These files need to be normalized to ensure the SQL queries meet syntax requirements. Since there is currently no dataset available for direct conversion from NL to SQL in the LCA domain, GPT-4.0 was used to understand the SQL statements and generate corresponding NL queries. By integrating these steps, the dataset for evaluation is ultimately constructed.

3.3. Automated report generation

In this paper, 10 LCIA result datasets from different scenarios were collected and analyzed. LLM agent first reads the datasets, analyzes the carbon footprint of the products based on the prompt provided by the user, and generates visual charts based on the analysis results. For different datasets, users can put forward specific analysis requirements, and LLM agent will automatically perform the corresponding data analysis.

The product carbon footprint report template from the standard (Standardization, 2018) was used as the basis for the format of the report generated by the LLMs. However, different word processing software (e.g. Microsoft Word, Google Docs, LibreOffice) may render and parse the same .docx file differently. This compatibility issue makes it challenging to generate Word files with consistent, cross-platform performance, which complicates LLMs' handling of formatting issues in .docx files. Given the LLMs' strong performance with documents in coded formats, the report template is converted into Markdown format. Markdown is a lightweight markup language that allows formatting elements to be added to plain text. The LLMs excel at parsing and generating content in Markdown syntax. The formatted report templates are shown in the Appendix D. The user can modify the LLMs' output, and the LLMs will provide feedback on any missing information in the report. Through continuous refinement and supplementation, a finalized product carbon footprint report is generated. Experiments were carried out on a dataset for the production of a 100 kg "lithium-sulfur battery" in Appendix E.

4. Evaluation

In Section 2.1.4, a QA dataset applicable to the LCA domain was constructed to evaluate the effectiveness of Chat-LCA. After that, comparative experiments have been conducted on GLM-4, Ernie-3.5-8K, Qwen1.5-72b, and DeepSeek-R1, both with and without RAG. The performance of various embedding models and retrievers was also evaluated. Subsequently, the Execution Accuracy in Text2SQL was used to evaluate the results of LLM-based fast retrieval of LCI databases. Finally, methods including qualitative and quantitative indicators were proposed to evaluate the reliability of LLMs in analyzing and generating reports for LCA data, so that a comprehensive assessment of the quality of automated product carbon footprint report generation was provided.

[3] https://huggingface.co/aspire/acge_text_embedding.
[4] https://www.trychroma.com/.
[5] https://www.gislca.com/.

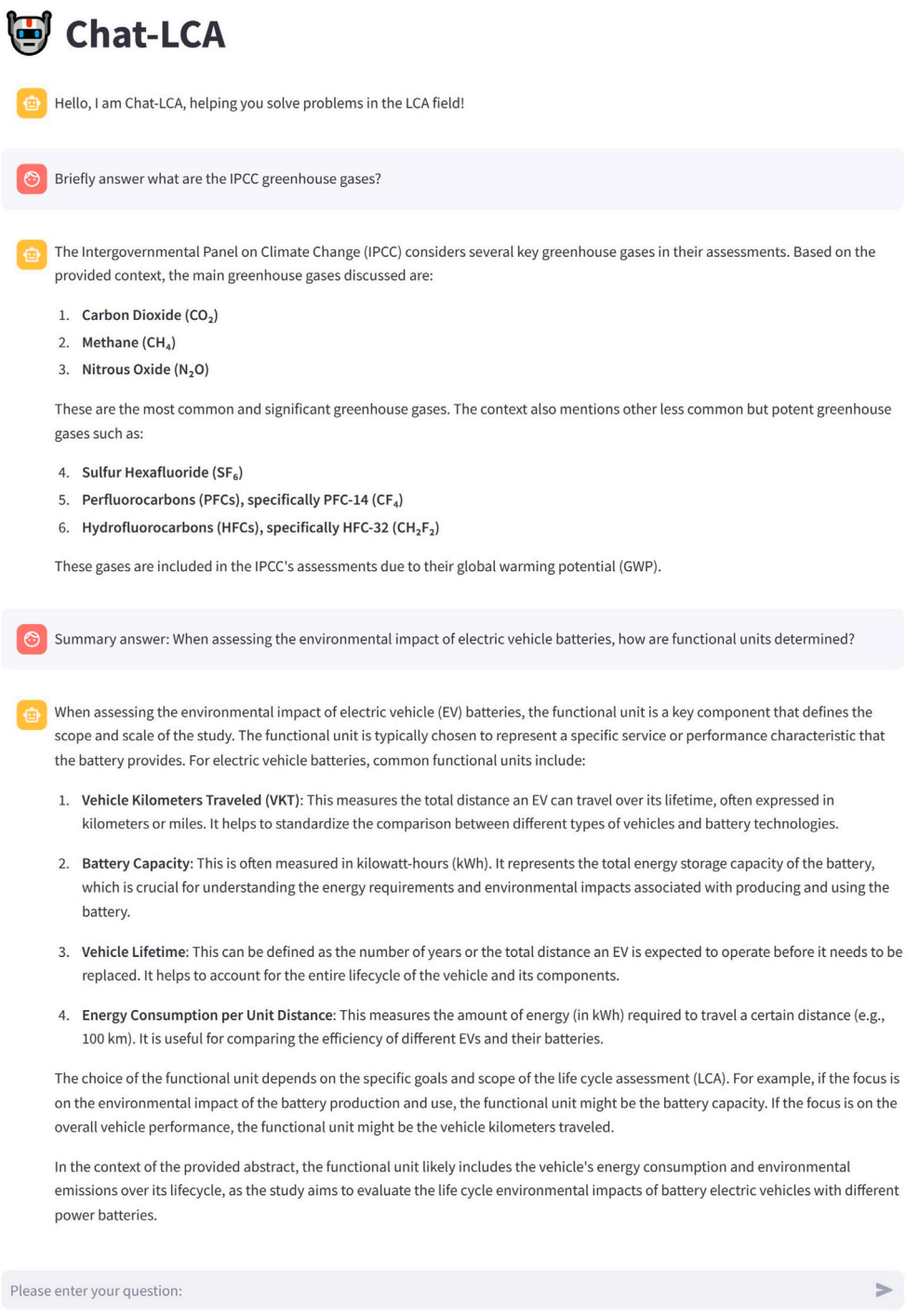


Fig. 8. Chat-LCA interactive interface.

4.1. Evaluation of LCA's professional QA

The QA dataset was generated based on authoritative LCA scientific and technical papers. In order to evaluate the effectiveness of the LLMs in answering questions, BERTScore (Zhang et al., 2019), cosine similarity and Rouge (Lin, 2004) are chosen as the evaluation metrics.

4.1.1. Metrics design

Cosine similarity can measure the similarity between two text vectors and evaluate the quality between the candidate answers obtained by LLMs on the LCA questions and reference answers. The pre-trained text embedding model is used to transform the reference answers and LLM-generated answers into text vectors and denote them as X and $\hat{X}$, respectively, where $X = (x_1, x_2, \ldots, x_n)$, $\hat{X} = (\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_m)$. The cosine similarity is computed as shown in formula (3), and if the vectors are normalized, the computation formula is $x_i^T \cdot \hat{x}_j$.

$$\text{cosine}(x_i^T, \hat{x}_j) = \frac{x_i^T \cdot \hat{x}_j}{\|x_i^T\| \|\hat{x}_j\|} \tag{3}$$

BERTScore measures semantic similarity using deep learning models, which consists of three components: recall, precision and F-1 score. BERTScore can more accurately assess the semantic consistency and contextual relevance of text, and is suitable for complex or diverse generated text. The complete score is calculated by matching each token in X to a token in $\hat{X}$ for recall, and each token in $\hat{X}$ to a token in X for precision. Then the recall and precision are combined to calculate the

Table 3
BERTScore on the LCA QA dataset.

	RAG	LTRK	EPEG	RED	CI	NMPI	CRMCPMI	CUWRI	AFAHFI	EHPSI	AMI
GLM-4	w/o	0.80	0.80	0.76	0.78	0.78	0.77	0.76	0.74	0.76	0.80
	w	0.81	0.82	0.79	0.81	0.80	0.79	0.80	0.80	0.80	0.82
Ernie-3.5-8K	w/o	0.81	0.80	0.76	0.78	0.77	0.76	0.77	0.76	0.77	0.81
	w	0.82	0.83	0.79	0.82	0.80	0.81	0.81	0.80	0.81	0.83
Qwen1.5-72b	w/o	0.81	0.79	0.76	0.77	0.76	0.75	0.75	0.73	0.74	0.78
	w	0.83	0.83	0.80	0.82	0.83	0.83	0.82	0.82	0.81	**0.85**
DeepSeek-R1	w/o	0.82	0.82	0.78	0.81	0.80	0.80	0.79	0.78	0.79	0.83
	w	**0.83**	**0.83**	**0.81**	**0.83**	**0.83**	**0.84**	**0.83**	**0.82**	**0.82**	0.84

Table 4
Cosine similarity on the LCA QA dataset.

	RAG	LTRK	EPEG	RED	CI	NMPI	CRMCPMI	CUWRI	AFAHFI	EHPSI	AMI
GLM-4	w/o	0.73	0.73	0.71	0.74	0.75	0.75	0.73	0.72	0.74	0.75
	w	0.75	0.76	0.73	0.77	0.76	0.76	0.77	0.77	0.79	0.79
Ernie-3.5-8K	0.73	0.73	0.70	0.72	0.73	0.74	0.73	0.73	0.74	0.76	
	w	0.76	0.77	0.74	0.78	0.77	0.78	0.78	0.77	0.80	0.79
Qwen1.5-72b	w/o	0.72	0.71	0.69	0.71	0.71	0.72	0.72	0.70	0.72	0.72
	w	0.76	0.77	0.74	0.77	0.79	0.81	0.78	0.79	0.79	0.80
DeepSeek-R1	w/o	0.75	0.75	0.73	0.77	0.76	0.77	0.76	0.75	0.76	0.78
	w	**0.78**	**0.79**	**0.76**	**0.80**	**0.80**	**0.82**	**0.81**	**0.79**	**0.81**	**0.81**

F-1 score. The recall, precision and F-1 score between gold answer X and candidate answer $\hat{X}$ are followed formula (4), (5), (6) respectively. The recall value of BERTScore is used as the evaluation metric.

$$R_{BERT} = \frac{1}{|X|} \sum_{x_i \in X} \max_{\hat{x}_j \in \hat{X}} x_i^T \hat{x}_j \tag{4}$$

$$P_{BERT} = \frac{1}{|\hat{X}|} \sum_{\hat{x}_j \in \hat{X}} \max_{x_i \in x} x_i^T \hat{x}_j \tag{5}$$

$$F_{BERT} = \frac{2P_{BERT} \cdot R_{BERT}}{P_{BERT} + R_{BERT}} \tag{6}$$

The Rouge metric is calculated by comparing the candidate answers generated by LLMs with the reference answers to get the corresponding scores (Lin, 2004). In this paper, the recall in the Rouge-L metric based on the longest common subsequence (LCS) is used to evaluate the quality of the generated answers. For text sequences S and T of length u and v, respectively, the Rouge-L metric can be computed by the formula (7).

$$R_{LCS} = \frac{LCS(S,T)}{u} \tag{7}$$

where $LCS(S,T)$ is the longest common subsequence of S and T.

All of the above indicators are in the range of 0–1, the closer to 1 the more accurate the answer is.

4.1.2. Metrics analysis

Through experiments, the results shown in Tables 3, 4, 5 were obtained. Across three evaluation metrics, LLMs enhanced with RAG consistently surpassed baseline LLMs without RAG. Specifically, DeepSeek-R1 with RAG exhibited superior performance on BERTScore and cosine similarity, while Qwen1.5-72b with RAG delivered the optimal results on the Rouge metric. BERTScore leverages deep learning to assess semantic similarity by integrating recall, precision, and F1 score, and is well-suited for evaluating a model's ability to comprehend and respond to complex LCA-related concepts, thus LLMs tend to perform well on this metric. In contrast, Rouge focuses on character-level overlap via LCS, emphasizing surface-level similarity rather than semantic understanding, which may undervalue the strengths of LLMs in semantic reasoning within the LCA context.

LLMs with RAG outperforms other LLMs without RAG on three metrics, indicating superior semantic understanding and matching. This can be attributed to the incorporation of the RAG technique, which supplements answers by retrieving relevant documents, allowing the model to better understand LCA domain-specific terms and complex concepts. The results show that LLMs with RAG not only understands the semantics of the question but also approximates the reference answer in linguistic presentation. When responding to questions, LLMs with RAG provides precise answers while covering more details related to the questions. The RAG that extracts contextual information from relevant documents, enables the LLMs to incorporate more critical data into its answers. It is particularly beneficial in the complex LCA domain, allowing the model to address multidimensional aspects of the information.

The average scores of various metrics for different LLMs with RAG on QA datasets were calculated shown in Table 6. Analysis indicates that DeepSeek-R1 with RAG demonstrates the best overall performance. Therefore, DeepSeek-R1 was chosen as the base model for Chat-LCA. Based on this, the impact of different text embedding models and retrievers on performance are further compared.

Table 7 presents the BERTScore evaluation results of the DeepSeek-R1 with RAG on the QA dataset when using three different embedding models in combination with two retrieval methods: MMR and BM25. Cosine similarity and Rouge score results are detailed in Appendix F. It can be observed that in the Chinese knowledge base environment, the combination of the acge_text_embedding and MMR achieved the best performance.

4.2. Analysis of retrieval on LCI database

Execution Accuracy (EX) (Qin et al., 2022) is used to evaluate the accuracy of LCI data obtained by NL interaction with databases. EX is computed by executing the outputs of real and predicted SQL queries based on the database. The predicted query is considered correct only if the execution of the predicted SQL query $\hat{V}$ and the true SQL query V have the same execution result, as shown in formula (8):

$$\text{score}(\hat{V}, V) = \begin{cases} 0, & \hat{V} \neq V \\ 1, & \hat{V} = V \end{cases} \tag{8}$$

EX is calculated following formula (9), where K is the number of SQL queries in the test dataset:

$$\text{EX} = \frac{\sum_{k=1}^{K} \text{score}(\hat{V}_k, V_k)}{K} \tag{9}$$

EX is a suitable choice for evaluating Text2SQL tasks because it directly measures the correctness of SQL queries, which is critical in LCI database searches to ensure that users can obtain accurate data through

Table 5
Rouge on the LCA QA dataset.

	RAG	LTRK	EPEG	RED	CI	NMPI	CRMCPMI	CUWRI	AFAHFI	EHPSI	AMI
GLM-4	w/o	0.22	0.16	0.17	0.12	0.14	0.08	0.11	0.14	0.06	0.09
	w	0.31	0.21	0.21	0.16	0.22	0.18	0.20	0.27	0.19	0.22
Ernie-3.5-8K	w/o	0.30	0.20	0.20	0.15	0.17	0.13	0.14	0.18	0.11	0.17
	w	0.32	0.22	0.25	0.17	0.21	0.18	0.20	0.24	0.13	0.21
Qwen1.5-72b	w/o	0.27	0.16	0.16	0.10	0.12	0.08	0.10	0.13	0.06	0.12
	w	**0.39**	**0.33**	**0.33**	**0.30**	**0.41**	**0.36**	**0.29**	**0.35**	**0.32**	**0.35**
DeepSeek-R1	w/o	0.31	0.18	0.22	0.16	0.15	0.11	0.14	0.16	0.09	0.17
	w	0.34	0.21	0.24	0.18	0.23	0.19	0.21	0.26	0.15	0.23

Table 6
Metric average score on the LCA QA dataset.

LLMs with RAG	Bertscore	Cosine similarity	Rouge
GLM-4	0.81	0.76	0.22
Ernie-3.5-8K	0.81	0.77	0.21
Qwen1.5-72b	0.82	0.78	**0.34**
DeepSeek-R1	**0.83**	**0.80**	0.22

NL queries. The larger the value of EX, the higher the correctness of the predicted SQL statement and the more effective LLMs are.

The experiments compare the EX computation results of different models and methods, as shown in Table 8. In the evaluation of the LCI data retrieval experiment, the evaluation dataset was derived from query data generated during the use of the GIS-LCA platform. After data cleaning and denoising, a total of 66 data entries were obtained. Additional case studies can be found in Appendix G for further reference. The augmented-prompt method, combining CoT and CoC, significantly improves Text2SQL performance. DeepSeek-R1 with augmented prompts achieves 0.9692 accuracy, outperforming Few-shot DeepSeek-R1 and the base model. All models excel with this strategy, using CoT for reasoning and CoC for SQL generation. The base Qwen1.5-72b model performs worst, but augmented prompting with guidance and templates mitigates its limitations. This method generalizes well in few-shot scenarios and sets a new standard for prompt engineering in structured tasks.

This method is particularly practical for users with limited database management experience, especially LCA researchers who frequently query LCI databases. It simplifies database interactions, enabling quick searches without requiring proficiency in complex SQL syntax, thereby significantly improving query efficiency and result accuracy. By lowering the technical barrier, it allows researchers to focus more on analysis and decision-making.

4.3. Reliability of LCA report generation

To ensure the accuracy, completeness and expected outcomes of automated LCA report generation with data analysis, both qualitative and quantitative assessment metrics are developed. The automated process is then evaluated in detail using manual statistical methods.

4.3.1. Qualitative evaluation

Qualitative evaluation focuses on the content quality of the report and its relevance to the results of the LCA data analysis. Therefore, qualitative indicators include readability and relevance. Readability means that the automatically generated report should be easy to read, including smooth text expression and clear presentation of charts and tables. A ten-point scale (1, 2, …, 10) is used to rate readability, with higher scores indicating better readability. Relevance assesses whether the report focuses on the core data analyzed by LCA, avoiding irrelevant content. Relevance is categorized into five levels (strong, stronger, moderate, weaker, weak) with corresponding scores of (5, 4, 3, 2, 1).

Currently, LLMs can be applied as judges in multiple fields (Li et al., 2024; Gu et al., 2024). Therefore, the readability and relevance evaluations of the ten reports are conducted using GPT-4.0 instead of manual assessment. This approach reduces subjective errors associated with manual evaluation, and the analysis is repeated three times to minimize random errors. The average of the three assessments is taken as the final score, and the overall average readability and relevance scores for the ten reports are calculated as the result of the LLMs shown in Table 9.

The results show that the automatically generated reports perform well in terms of readability, with largely fluent text and clearer presentation of charts and tables. Chat-LCA received the highest score 8.39, indicating that LLMs learn to standardize the readability of LCA domain reports through the knowledge base. The LLMs' performance in the experiment range from 7 to 8 points, indicating that there are still some reports that need to be improved in terms of textual expression or visual presentation. Regarding relevance, the results range from 4 to 5 points, suggesting that the LLMs effectively focus on the core data analyzed by LCA and avoid generating unrelated content. Since more capable of generating content related to the LCA analysis of the report topic, the Chat-LCA performs best. Some reports even achieves a relevance score of 5, indicating that the report content is highly consistent with the analysis results. Reports scoring 4 points show a certain degree of relevance but contained some content that is not entirely related to the analysis topic.

4.3.2. Quantitative evaluation

Quantitative evaluation focuses on the elements to be filled in the statistical report. The quality of the generated report is evaluated by the fill-in completeness rate $Rate_{fill}$, the fill-in accuracy rate $Rate_{acc}$ and the number of key data items filled in, denoted as N_{cover}, which are calculated with the formula (10), (11). The report is then analyzed to determine whether the key data items are correctly labeled. In this study the total number of key data items in the report is defined as 4, which include carbon footprints at each stage of the product, the percentage of carbon footprints at each stage, the percentage of carbon footprints in the histogram or pie chart, and the correctness of the advice given for decision making.

$$Rate_{fill} = \frac{B}{C} \tag{10}$$

$$Rate_{acc} = \frac{B_{acc}}{S - Np} \tag{11}$$

where $S = 28$ is the total number of blank entries in the report template and is a fixed value, B is the number of blank entries in the report, B_{acc} is the number of blank entries in the report that are filled in accurately, Np is the number of information items not provided. Larger values of the above three metrics indicate higher quality of carbon footprint reports generated by LLMs.

$Rate_{fill}$ is designed to assess the effectiveness of the LLMs in completing the report text. But it is only a measure of whether the report is filled out completely. The reason for the lower $Rate_{fill}$ of Chat-LCA is the lack of learning about unknown blank content in the report. Based on the RAG and zero-shot learning, the Chat-LCA does not take the liberty of writing non-compliant content, thus some content is not filled in. Generally speaking, analysis of the evaluation results shown in Table 9 revealed that the fill-in-completeness rate ranged from 0.9416

Table 7
BERTScore on different embedding models and retrievers.

Embedding	Retrieval	LTRK	EPEG	RED	CI	NMPI	CRMCPMI	CUWRI	AFAHFI	EHPSI	AMI
acge_text_embedding	MMR	**0.83**	**0.83**	**0.81**	**0.83**	**0.83**	**0.84**	**0.83**	**0.82**	**0.82**	**0.84**
	BM25	0.82	0.82	0.79	0.81	0.81	0.79	0.79	0.76	0.79	0.82
bge-large-zh-v1.5	MMR	0.83	0.83	0.80	0.83	0.82	0.83	0.83	0.81	0.82	0.84
	BM25	0.82	0.83	0.79	0.82	0.81	0.80	0.80	0.77	0.78	0.83
gte-large-zh	MMR	0.83	0.83	0.81	0.83	0.83	0.83	0.82	0.80	0.82	0.84
	BM25	0.82	0.83	0.79	0.81	0.81	0.81	0.80	0.77	0.79	0.83

Table 8
Evaluation results of prompt enhanced database search.

LLMs	EX
Ernie-3.5-8k	0.8769
Qwen1.5-72b	0.8615
DeepSeek-R1	0.8923
Few-shot Ernie-3.5-8k	0.9077
Few-shot Qwen1.5-72b	0.9385
Few-shot DeepSeek-R1	0.9538
Augmented-prompt-Ernie-3.5-8k	0.9231
Augmented-prompt-Qwen1.5-72b	0.9538
Augmented-prompt-DeepSeek-R1	**0.9692**

Table 9
Evaluation of report generation.

Model	Readability	Relevance	$Rate_{fill}$	$Rate_{acc}$	N_{cover}
GLM-4	7.97	4.73	0.9481	0.9428	3.6
Ernie-3.5-8K	8.03	4.76	0.9416	0.9425	3.7
Qwen1.5-72b	7.60	4.43	**0.9679**	0.9528	3.5
DeepSeek-R1	7.91	3.85	0.9351	0.9646	3.5
Chat-LCA	**8.42**	**4.82**	0.9481	**0.9832**	**3.8**

to 0.9679, indicating a high average completion rate. It is demonstrated that the report generation process covers most of the template's blank items.

With the same information provided by the users, Chat-LCA achieves the highest $Rate_{acc}$ of 0.9832, which is significantly higher than other LLMs. It is attributed to the fact that Chat-LCA can access specialized data through the LCA knowledge base, enhancing the accuracy of specialized knowledge in report generation. However, a small number of elements are still filled in incorrectly, mainly due to inaccurate or missing information provided by users. Therefore, both a professional LLM and the users' cooperation in providing accurate and complete product information are essential for LLMs to effectively generate reports automatically.

The number of key data items filled in serves as an indicator to assess whether the system correctly covers the key data items. The difference in the number of keywords filled in by the five LLMs is minimal, indicating that all the LLMs are capable of integrating the results of the data analysis into the report with high quality.

4.4. Comparative analysis and discussion

The proposed method applies LLMs throughout the LCA process, advancing knowledge summarization, data retrieval, and automated report generation. Unlike prior studies that fine-tuned LLMs for specific domains like green hydrogen (Chen et al., 2024), this approach provides a generalized framework across industries, achieving a BERTScore of 0.85 in QA tasks. Compared to domain-specific methods, the RAG-based Chat-LCA offers greater adaptability to diverse scenarios. While some work focused on processing LCI databases, they mainly addressed data collection and lacked quantitative evaluation (Tu et al., 2024). This paper emphasizes accurate LCI data retrieval and introduces EX metrics for performance assessment. For carbon footprint reporting, where existing research is sparse, we propose customized indicators to evaluate report quality.

5. Conclusion and future work

This study proposes and validates a method that integrates LLMs' domain-specific techniques to enhance the efficiency of LCA research. The method includes LCA professional QA, LCI data retrieval, and automated report generation. Adopting RAG to build an LLM in LCA specialized fields allows for the coverage of LCA-related knowledge across multiple industries. It helps LCA practitioners quickly acquire relevant knowledge and experience, alleviating the hallucinations of LLMs in LCA. The integrated approach of Text2SQL with hybrid augmented prompts facilitates NL retrieval to LCI databases. The barrier for professional searches in databases is reduced, which provides a foundation for automated data management in the LCA field. Automated report generation on LCIA results are achieved through code-interpreter agent combined with the zero-shot learning. The results of the study also show the potential of the LLMs to be able to be integrated with LCA domain tools, such as the GIS-LCA platform. The application of LLMs in LCA not only advances the innovative integration of cutting-edge technologies but also provides novel solutions for processing and analyzing complex environmental issues.

While integrating LLMs into LCA offers clear advantages, several challenges remain. To enhance response quality, domain-specific training data should be developed to fine-tune LLMs for LCA applications. Improving the interpretability of model outputs is also critical for building user trust and supporting decision-making, yet suitable evaluation benchmarks for the LCA domain are lacking. The evaluation dataset in Section 4.2 requires further expansion in terms of its generalizability. Therefore, it is a research focus to construct an evaluation dataset suitable for the LCA field using natural language to retrieve LCI. The LCI modeling phase faces issues such as mismatched background data and incomplete foreground data, which could benefit from advanced LLMs and deep learning techniques to improve data management. Data privacy and security also require attention, with mechanisms needed to prevent information leakage. In the future, LLMs applications in LCA should focus on deeper integration with industry workflows. LLM-driven platforms can facilitate NL interaction with LCA tools and databases, improving accessibility for practitioners. Ultimately, embedding LLMs at the system level will be essential to realizing their full potential in solving complex environmental challenges.

CRediT authorship contribution statement

Xiaojun Zhang: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Xiang Guo:** Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Jinghao Zhao:** Writing – review & editing, Supervision, Conceptualization. **Jie Xiong:** Resources, Data curation. **Yajun Tian:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Experimental programming tools

The experimental operating system used in this paper is Ubuntu 20.04.4, and the software tools include Pycharm and VScode. This paper uses Python as the experimental programming language and makes full use of tool libraries widely used in LLMs development, such as `langchain`, `transformers`, and `torch`. The above tools can be installed using the `pip install *` command.

Appendix B. Table of abbreviations

Abbreviation	
AFAHFI	Agriculture, forestry, animal husbandry and fishery industry
AMI	Automobile manufacturing industry
CI	Construction industry
CRMCPMI	Chemical raw materials and chemical products manufacturing industry
CUWRI	Comprehensive utilization of waste resources industry
EHPSI	Electricity, heat production and supply industry
EPEGI	Ecological protection and environmental governance industry
LTRK	LCA theory and related knowledge
NMPI	Non-metallic mineral products industry
RED	Research and experimental development

Appendix C. Prompt for LLMs enhanced LCI database retrieval

C.1. Instruction

You are a SQL query generation assistant. Given a NL query and a database schema, you process it internally as follows, but only output the final SQL statement:
1. Chain of Thought Reasoning (internal execution):
- Analyze the natural language query step by step to clarify the following:
- Involved tables and aliases.
- Columns that need to be selected.
- Filter conditions (WHERE clause).
- Whether aggregation, join, grouping or sorting is required.
- Ensure that the reasoning is clear and avoid semantic ambiguity.

2. Chain of Code Decomposition (internal execution):
- Based on CoT reasoning, generate a structured JSON intermediate representation, including:
- Tables
- Conditions
- Aggregation
- Use JSON representation to ensure logical modularity.

3. SQL Generation (internal execution):
- Generate SQL queries based on JSON representation.
- Use table aliases to ensure that the syntax is correct and complies with the database schema.

4. Validation (internal execution):
- Check whether the SQL complies with the database schema (whether the table and columns exist).
- Ensure that the SQL reflects the query intent and has no syntax errors.

C.2. Augmented prompt example

Given the provided life cycle inventory database schema information and a question, please write the corresponding SQL query statement for the question.
"Question":How many undeleted records are there in the database tb_process table?
"Table Information":[⋯].
"Foreign Keys":"".
Please start by outlining your thought process, and then output the final SQL in the following format:
Generated SQL:
‘‘‘
{sql}
‘‘‘

Appendix D. Carbon footprint study report template

Product Carbon Footprint Study Report (Template)

Basic Information

- Product Name: [1]
- Product Specification Model: [2]
- Producer Name: [3]
- Report No: [4]
- Reporting organization: (if any)
- Date:[5]

I. General Information

- Producer Name: [6]
- Address: [7]
- Legal representative:[8]
- Product Name:[9]
- Product Function: [10]
- Based on the standard: [11]

II. Quantitative Purpose

(Enter quantitative purpose here) [12]

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in ( ).[13]

2. System Boundary

(Requires user-supplied system boundary maps)[14]

3. Cut off Criteria

The rounding criteria used are based on ( ) and the rules are as follows:[15]

4. Time Scale

(Annual).[16]

IV. Inventory Analysis

1. Description of Data Sources

- Primary data: ( ) [17]
- Secondary data: ( ) [18]

2. Principles and Procedures for Allocation

- Basis of allocation: [19]
- Allocation Procedure: [20]
- Specific allocations: [21]

3. Data Quality Evaluation (optional)

(Complete the data quality assessment checklist; data quality can be evaluated in terms of both qualitative and quantitative aspects of the primary and secondary data used in the report in terms of data sources, completeness, data representativeness (temporal, geographic, technological) and accuracy.)[22]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

Generally choose the 100-year global warming potential (GWP) given by the Intergovernmental Panel on Climate Change (IPCC). [23]

2. Product Carbon Footprint Results Calculation

(fill in calculations here) [24]

Figure 1: Adding visualization results [25]

Table 1:Carbon emissions by life cycle stage [26]

Life cycle stage	Carbon footprint (kg-CO2e)	Percentage (%)
Raw Material Acquisition		
Production		
Distribution		
Use		
End of life		
Total		

VI. Interpretation of Results

1. Description of Results

The (fill in the full name of the product producer) company produces (fill in the name of the product evaluated, per functional unit of product) with a life cycle carbon footprint of () kg CO_2e from (fill in a life cycle stage) to (fill in a life cycle stage). The GHG emissions for each life cycle stage are shown in Table 1 and Figure 1.

2. Statement of Assumptions and Limitations

The assumptions and limitations related to scope, data selection, scenario setting, etc. are described in detail in the context of quantification. [27]

3. Suggestions for Improvement

(insert suggestions for improvement here) [28]

Note: [*] indicates the items of information to be filled in.

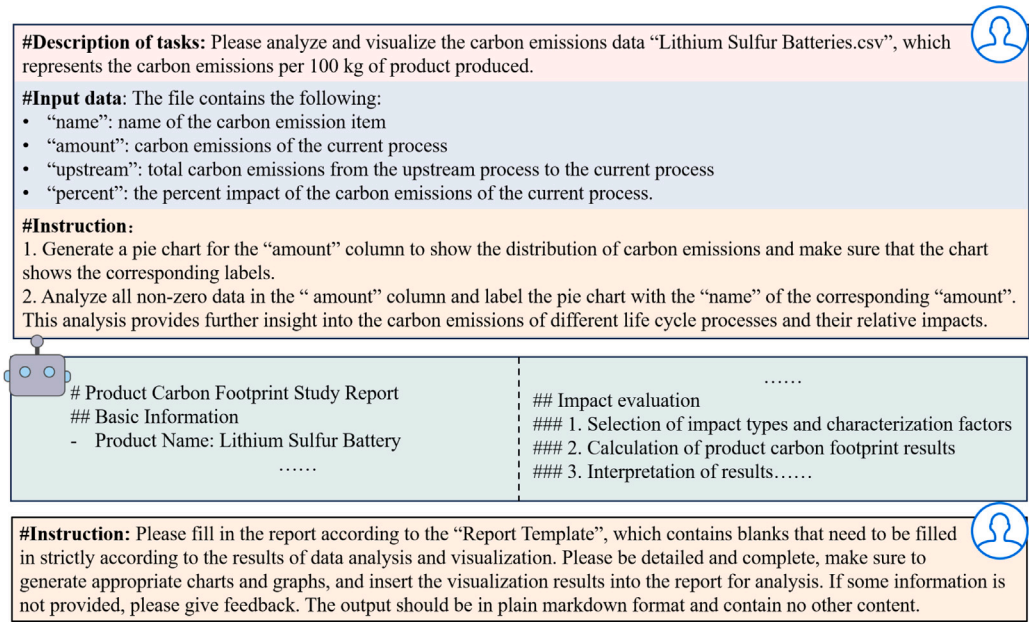


Fig. E.9. Example Prompt of "Lithium Sulfur Batteries" data analysis and report generation.

Appendix E. Carbon footprint study report of lithium sulfur battery

For example, consider the environmental impact data generated for the production of a 100 kg "lithium-sulfur battery" using data from the GIS-LCA platform. This data includes the material name, the amount of material required, carbon emissions from the upstream process, and the percentage of carbon emissions from the material. Automated data analysis and report generation are guided by the prompt shown in Fig. E.9, and the final report is shown in the follow. Missing information is added based on feedback from the LLMs, with the specific additions varying according to the data. Ultimately, the carbon footprint report for the "lithium-sulfur battery" is produced, covering sections such as basic information, overview, quantification purpose, scope, inventory analysis, impact evaluation, and interpretation of results, as specified in the template.

Product Carbon Footprint Study Report

Basic Information

- Product Name: Lithium Sulfur Battery
- Product Specification Model: [Information provided by the user]
- Producer Name: Product Company
- Report No: [Information provided by the user]
- Reporting organization: [Information provided by the user]
- Date:November 19, 2024

I. General Information

- Producer Name: Product Company
- Address: Product Company
- Legal representative: Zhang
- Product Name: lithium-sulfur battery
- Product Function: High energy density energy storage device for portable electronic devices, electric transportation, etc.
- Based on the standard: IPCC 2013 GWP 100a

II. Quantitative Purpose

The purpose of this quantification is to assess the greenhouse gas (GHG) emissions of lithium-sulfur batteries at various stages of production, distribution, use, and end-of-life disposal to determine the extent of their impact on climate change and to identify opportunity points for reducing their carbon footprint.

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in terms of Units of energy.

2. System Boundary

(User-supplied system boundary diagram is required): User-supplied information

3. Cut off Criteria

The trade-off criteria adopted are based on IPCC 2013 GWP 100a, and the specific rules are as follows:
- All toxic and hazardous substances should be listed; - All toxic and hazardous substances should be listed. Energy and raw material inputs in any form are listed; - Some raw material trimmings can be directly recycled and used as raw materials, so they can be ignored. - The whole process of paint production and application uses China's national power grid, and the level of enterprise safety production standardization meets the requirements of AQ/T9006.

4. Time Scale

Year 2023.

IV. Inventory Analysis

1. Description of Data Sources

- Primary data: GIS-LCA platform
- Secondary data: None

2. Principles and Procedures for Allocation

- Basis of allocation: Based on different stages of product life cycle.
- Allocation procedure: Allocation by analyzing data in CSV files.
- Specific allocation: See "Impact Assessment" below.

3. Data Quality Evaluation (optional)

The results of the data quality assessment indicate that the primary data used have a high degree of completeness and accuracy, and are representative of the carbon emissions of lithium-sulfur batteries throughout their life cycle.

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

The 100-year global warming potential (GWP) given by the Intergovernmental Panel on Climate Change (IPCC) is generally selected.

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of lithium-sulfur batteries produced by Product Company, per functional unit of product, from raw material acquisition to end of life is 30 223.8 kg CO2e. The GHG emissions at each life cycle stage are shown in Table 1 and Fig. 1.

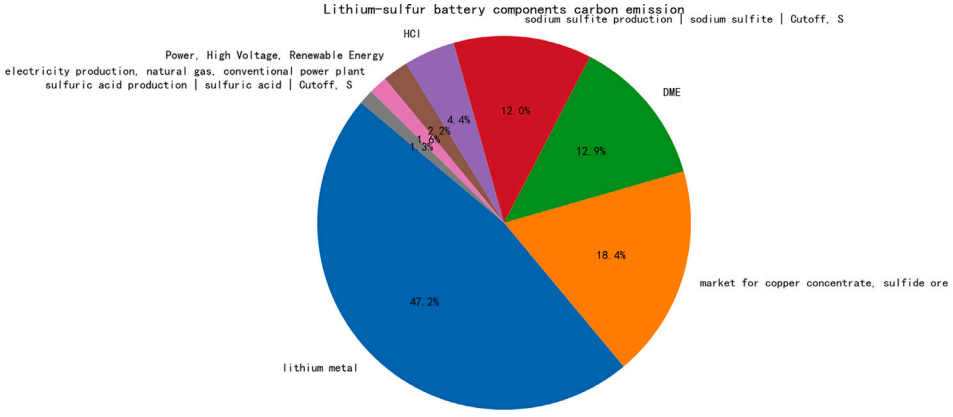


Table F.10
Cosine similarity on different embedding models and retrievers.

Embedding	Retrieval	LTRK	EPEG	RED	CI	NMPI	CRMCPMI	CUWRI	AFAHFI	EHPSI	AMI
acge_text_embedding	MMR	**0.78**	**0.79**	**0.76**	**0.80**	0.80	**0.82**	**0.81**	**0.79**	**0.81**	**0.81**
	BM25	0.76	0.76	0.74	0.77	0.78	0.76	0.76	0.74	0.77	0.78
bge-large-zh-v1.5	MMR	0.77	0.78	0.76	0.79	0.80	0.82	0.81	0.79	0.80	0.81
	BM25	0.76	0.76	0.74	0.77	0.77	0.77	0.76	0.75	0.76	0.79
gte-large-zh	MMR	0.78	0.78	0.76	0.80	**0.81**	0.81	0.80	0.78	0.80	0.81
	BM25	0.76	0.77	0.74	0.77	0.78	0.78	0.77	0.75	0.78	0.79

Table F.11
Rouge on different embedding models and retrievers.

Embedding	Retrieval	LTRK	EPEG	RED	CI	NMPI	CRMCPMI	CUWRI	AFAHFI	EHPSI	AMI
acge_text_embedding	MMR	**0.34**	**0.21**	**0.24**	**0.18**	**0.23**	**0.19**	**0.21**	**0.26**	0.15	**0.23**
	BM25	0.30	0.18	0.21	0.13	0.16	0.10	0.14	0.16	0.08	0.16
bge-large-zh-v1.5	MMR	0.32	0.21	0.25	0.17	0.24	0.18	0.21	0.24	0.16	0.24
	BM25	0.29	0.17	0.21	0.14	0.15	0.09	0.15	0.15	0.06	0.16
gte-large-zh	MMR	0.31	0.21	0.24	0.18	0.23	0.18	0.19	0.20	**0.18**	0.17
	BM25	0.30	0.19	0.22	0.14	0.17	0.11	0.15	0.15	0.10	0.16

Table 1:Carbon emissions by life cycle stage

Life cycle stage	**Carbon footprint (kg-CO2e)**	**Percentage (%)**
Raw Material Acquisition	13 328.5	47.2
Production	10 461.7	31.3
Distribution	3658.9	12.0
Use	1248.3	4.4
End of life	1526.4	5.1
Total	30 223.8	100

VI. Interpretation of Results

1. Description of Results

The life cycle carbon footprint of lithium-sulfur batteries manufactured by Products, per functional unit of product, from raw material acquisition to end-of-life is 30,223.8 kg CO2e. As seen in the table and figure above, the raw material acquisition stage accounts for the largest share of all life cycle stages, amounting to 47.2% of the total carbon emissions. The manufacturing stage follows with 31.3%, while the distribution, use, and end-of-life stages contribute 12.0%, 4.4%, and 5.1% of the carbon emissions, respectively.

2. Statement of Assumptions and Limitations

The data in this report are derived from measurements under specific conditions and therefore may not fully reflect the true carbon footprint of lithium-sulfur batteries in all cases. For example, different production processes and technology levels may result in variations in actual carbon emissions. In addition, for certain processes that are difficult to measure accurately (e.g., transportation distance), we have used an average estimate for the calculation.

3. Suggestions for Improvement

In order to reduce the carbon footprint of lithium-sulfur batteries, the following measures are recommended:
- Enhance the efficiency of the raw material procurement process to minimize unnecessary energy consumption;
- Promote the application of clean energy, especially increasing the proportion of renewable energy in the manufacturing process;
- Optimize packaging design to reduce resource waste during transportation;
- Strengthen waste management and increase the recycling rate of used batteries.

Appendix F. Comparative experimental results of embedding models and retrievers

See Tables F.10 and F.11.

Appendix G. Table of NL question and SQL query

NL question	SQL query
Count the number of records for each dataset_owner_id in the tb_process_doc table.	`SELECT dataset_owner_id, COUNT(*) AS record_count FROM tb_process_doc GROUP BY dataset_owner_id;`
Find all records in the tb_process table where update_time is after October 15, 2024.	`SELECT * FROM tb_process WHERE update_time > '2024-10-15';`
Find all records in the tb_process table where default_allocation_method is "economic allocation".	`SELECT * FROM tb_process WHERE default_allocation_method = 'economic allocation';`

Data availability

Data will be made available on request.

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